

Test Cases

Test Case ID	Input Scenario	Expected Output	Result
TC001	Generate MERN Stack setup on Ubuntu	Bash installation script generated successfully	Pass
TC002	Generate Python ML environment on macOS	Python, TensorFlow, PyTorch setup script generated	Pass
TC003	Generate Java Spring Boot setup on Windows	PowerShell installation script generated	Pass
TC004	Generate Docker & Kubernetes environment	Docker and Kubernetes installation script generated	Pass
TC005	Request Docker Compose configuration	Docker Compose YAML file generated	Pass
TC006	Generate script with missing OS information	System requests additional details	Pass
TC007	Generate script with unsupported software	Validation error displayed	Pass
TC008	Security scan of generated script	Dangerous commands detected and flagged	Pass
TC009	Download generated script	Script downloaded successfully	Pass
TC010	View previously generated script	Script loaded from history successfully	Pass
TC011	User Login	JWT token generated successfully	Pass
TC012	User Registration	New user account created successfully	Pass
TC013	Search Script History	Relevant scripts displayed	Pass
TC014	Delete Script History Record	Script removed successfully	Pass
TC015	Multi-Agent Workflow Execution	All 7 agents executed successfully	Pass

Functional Testing

User Authentication

- Verify user registration works correctly.
- Verify user login generates JWT token.
- Verify unauthorized access is blocked.
- Verify logout functionality works.

Script Generation

- Verify user prompt is processed correctly.
 - Verify operating system detection works.
 - Verify dependency analysis is performed.
 - Verify script generation output is produced.
 - Verify documentation report is generated.
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Multi-Agent Workflow

- Verify Coordinator Agent executes.
 - Verify Dependency Analysis Agent executes.
 - Verify OS Detection Agent executes.
 - Verify Security Validation Agent executes.
 - Verify Compatibility Agent executes.
 - Verify Script Generator Agent executes.
 - Verify Report Generation Agent executes.
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Security Testing

- Verify dangerous commands are detected.
 - Verify unsafe scripts are flagged.
 - Verify input sanitization works correctly.
 - Verify API endpoints are protected.
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Script History Testing

- Verify generated scripts are saved.
 - Verify history records are retrieved.
 - Verify search functionality works.
 - Verify deletion functionality works.
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Download Testing

- Verify Bash (.sh) file downloads correctly.
 - Verify PowerShell (.ps1) file downloads correctly.
 - Verify Docker Compose (.yaml) file downloads correctly.
 - Verify Markdown documentation downloads correctly.
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Dashboard Testing

- Verify dashboard loads successfully.
 - Verify script generation form works.
 - Verify agent execution logs display correctly.
 - Verify recent scripts section loads.
 - Verify navigation menu works.
 - Verify user profile page loads.
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Database Testing

- Verify SQLite database is created.
 - Verify generated scripts are stored.
 - Verify user information is stored.
 - Verify history records are retrieved correctly.
 - Verify deleted records are removed.
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API Testing

- Verify /api/auth/signup works correctly.
 - Verify /api/auth/login works correctly.
 - Verify /api/generate-script returns valid output.
 - Verify /api/history returns stored scripts.
 - Verify /api/download-script/{id} downloads files.
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Performance Testing

- Generate 10 scripts within 10 seconds.
 - Generate 50 scripts without server crash.
 - Database insertion success rate: 100%.
 - Simultaneous user requests handled successfully.
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Error Handling Testing

- Missing operating system input.
- Missing software stack input.
- Invalid request format.
- Unsupported operating system.

- Database connection failure.
- Gemini API unavailable.
- Invalid JWT token.
- Empty user prompt.